[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

GRINDING MACHINE, SURFACE, RECIPROCATING-TABLE, HORIZONTAL-SPINDLE, POWER-FEED

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers floor-mounted, power-feed, horizontal-spindle, reciprocating-table, surface grinding machines. The machines are intended for rough and precision finish surface grinding and slot or form grinding operations on a wide variety of materials.

2. CLASSIFICATION. The grinding machines shall be of the following sizes. The size to be furnished shall be as specified (see 7.2(b)).

Size: See table I

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3415

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TABLE I. Machine sizes.

Size	Worktable dimensions	
	Width, inches	Length, inches
1	6	18
2	8	18
3	8	24
4	10	20
5	12	24
6	12	36
7	16	24
8	16	28
9	16	36
10	16	40
11	20	24
12	20	40
13	20	60
14	20	80
15	24	36
16	24	60
17	24	80
18	24	120

Note 1: Machine size and capacity shall not be less than the stated requirements selected from tables I and II for the class, swing, and size specified. The ranges of size and capacity shown in tables I and II imply that the machine offered shall be equal to or larger than the size and capacity stated but smaller than the next available size. When the largest machine shown in tables I and II is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent.

3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.1.1 Identification of axes. The grinding machine shall be provided with three mutually perpendicular axes of motion as defined in American National Standards Institute/Electronic Industries Association (ANSI/EIA) 267-C.

3.2 Components. The grinding machine shall include, as a minimum, the following components:

3.2.1 Base. The base shall support the other elements of the machine and shall incorporate materials, shape, and mass to minimize adverse vibration and deflection. The base shall provide a means for leveling and anchoring the machine. The base shall be of sufficient area and design to ensure that the center of gravity of the worktable cannot extend beyond the periphery of the base.

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3.2.2 Machineways. Unless otherwise specified (see 7.2(c)), machineways shall be of the design considered by the manufacturer as standard for the machine. All machineways shall be provided with protective means to keep chips, coolant, and other debris from bearing surfaces.

3.2.3 Worktable. The worktable shall be designed to minimize distortion under varying work loads. The worktable shall be provided with a means for powered positioning and reciprocating motion. A means shall be provided to prevent overtravel. Unless otherwise specified (see 7.2(d)), table travel shall be equal to or greater than the table dimensions. Unless otherwise specified (see 7.2(e)), the worktable shall have T-slots for mounting work-holding fixtures and magnetic chucks. The number and size of T-slots shall be provided as specified (see 7.2(f)). When T-slots are provided, the dimensions of the T-slots shall conform to the requirements of ANSI B5.1. A deflector shall be provided to catch the swarf thrown by the wheel during grinding.

3.2.4 Magnetic chuck. If required, a permanent magnetic or electromagnetic chuck shall be provided as specified (see 7.2(g)). Unless otherwise specified (see 7.2(h)), the working surface area of the chuck shall be the same as that of the worktable. When a magnetic chuck is provided, its control shall be interlocked with the control for the grinding wheel spindle motor to prevent spindle rotation without an activated chuck. The chuck shall be sealed for protection from coolant and grinding swarf.

3.2.5 Wheelhead assembly. As a minimum, the wheelhead assembly shall consist of the following:

a. A supporting frame provided with a means for powered positioning and travel.
b. A grinding wheel spindle and spindle drive motor mounted on the supporting frame.
c. A protective housing to encompass the components of the assembly.
d. Grinding wheels(s) and appropriate safety cover.
e. A means to prevent wheelhead assembly overtravel.

3.2.6 Grinding wheel spindle. The grinding wheel spindle shall be supported and held in axial and radial alignment by precision antifriction or low-friction bearings and be dynamically balanced to provide vibration-free operation at all spindle speeds. Continuous or permanent lubrication of the bearings shall be provided. Unless otherwise specified (see 7.2(i)), the spindle drive, horsepower, and speed shall be that considered by the manufacturer as standard for the grinding wheel(s) provided. When required, the spindle shall accept premounted and balanced grinding wheels as specified (see 7.2(j)). Unless otherwise specified (see 7.2(k)), the maximum height of the spindle above the worktable shall be that considered by the manufacturer as standard.

3.2.7 Grinding wheel(s). Unless otherwise specified (see 7.2(l)), the grinding wheel type(s) and size(s) shall be that considered by the manufacturer as standard. The wheel(s) shall be marked in accordance with ANSI B74.13 and conform to the safety requirements of ANSI B7.1.

3.2.8 Grinding wheel dresser. Unless otherwise specified (see 7.2(m)), a grinding wheel dresser with at least a one-carat diamond shall be provided to true the face of the grinding wheel. The dresser shall be capable of truing the grinding wheel to its full usable depth. Dressing shall not require removal of the workpiece.

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3.2.9 Grinding wheel forming device. When specified (see 7.2(n)), a power-driven, grinding wheel forming device shall be provided to shape the face of the grinding wheel. The forming device shall be mountable on the wheelhead assembly or worktable outside the work zone and accessible for changing form wheels. The forming device shall be capable of forming the grinding wheel to its full usable depth.

3.2.10 Dials and indicating devices. Unless otherwise specified (see 7.2(o)), all dials and indicating devices used to indicate machine axis position or movement shall be graduated in increments not greater than 0.0001 inch. Dial and handwheel circumjacent scales shall be permanently and legibly engraved or etched on a nonglare background. All feed dials shall have independent zero adjustment and be calibrated contiguously over the full range of movement.

3.2.10.1 Digital readout. When specified (see 7.2(p)), a digital readout system shall be provided. The system shall include all feedback devices and components necessary to provide a digital readout of position for the full working range of all axes. All components of the digital readout system shall be protected from damage caused by the normal grinding environment of vibration, coolant splash, grinding swarf or dust, and electric power fluctuations. Unless otherwise specified (see 7.2(q)), the digital readout system shall meet the requirements shown in table II.

TABLE II. Digital readout system requirements.

Parameter	Requirement
Resolution	0.00001 inch
Accuracy	±0.00005 inch
Repeatability	±0.00005 inch

3.2.11 Controls. All controls shall be convenient to the operator and positioned to reduce operator fatigue. Controls shall be functionally grouped and clearly identified. Status indicators, meters, and displays relevant to the operation of the machine shall be easily visible to the operator. All components of the control system shall be protected from damage caused by the normal grinding environment of vibration, coolant splash, grinding swarf or dust, and electric power fluctuations.

3.2.11.1 Control panel. Unless otherwise specified (see 7.2(r)), the control panel shall include, as a minimum, the following manual controls and switches:

a. Emergency stop switch.
b. Spindle on/off power switch, and when applicable, speed control adjustment.
c. Coolant on/off power switch and flow rate control adjustment.
d. Dust extractor on/off power switch.
e. Wet/dry grinding selector switch.
f. Rapid positioning control.
g. Sparkout timer and/or number of passes (automatic down feed cycle).
h. Course and fine feed (automatic down feed cycle).

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3.2.11.2 Handwheel feed controls. Unless otherwise specified (see 7.2(s)), the control system shall include, as a minimum, the following handwheel feed control functions:

a. Grinding wheel down feed.
b. Worktable longitudinal feed.
c. Cross feed.

3.2.11.3 Power feed controls. Unless otherwise specified (see 7.2(t)), the control system shall include, as a minimum, the following power feed control functions:

a. Wheelhead down feed speed and incremental input positioning.
b. Worktable longitudinal reciprocation speed and adjustable stop positioning.
c. Cross feed speed and adjustable stop positioning.

3.2.11.4 Automatic down feed cycle control. When specified (see 7.2(u)), automatic down feed cycle control shall be provided and shall operate in accordance with the following minimum requirements. The down feed cycle shall automatically engage worktable longitudinal and cross feed travels and lower the grinding wheel based on predetermined input stops at each cross feed reversal until spark out of the workpiece is complete. Spark out shall be set based on the time or number of passes selected prior to engaging automatic cycle. Upon completion of spark out, the grinding wheel shall retract to the starting position, worktable longitudinal and cross feed movement shall stop, and grinding wheel rotation shall stop. The automatic down feed cycle shall be capable of repeating the previous cycle without having to re-enter the input parameters.

3.2.11.5 Control interlock. The controls for the coolant-circulating and dust-extractor systems shall be interlocked to provide independent operation and prevent simultaneous operation of these systems. When a magnetic chuck is provided, the controls for the grinding wheel and chuck shall be interlocked to provide independent activation of the chuck and prevent operation of the grinding wheel without activation of the chuck.

3.2.12 Dust extractor system. Unless otherwise specified (see 7.2(v)), a dust extractor system shall be provided for dry grinding. The system shall collect dust generated during dry grinding and transport it to a filter for removal. When the grinding machine is intended for connection to a central dust collection system, the interface between the central system and the machine shall be provided as specified (see 7.2(w)).

3.2.13 Lubrication system. When specified (see 7.2(x)), the grinding machine shall be equipped with an automatic lubricating system to direct oil flow from the reservoir to all machine components requiring continuous lubrication. A means shall be provided for determining oil level in the reservoir. The automatic lubricating system shall include a cleanable or replaceable filter. Upon loss of pressure in the automatic lubrication system, the grinding machine shall automatically stop. Machine parts requiring periodic manual lubrication shall be readily accessible through doors or removable panels.

3.2.14 Coolant system. Unless otherwise specified (see 7.2(y)), the grinding wheel shall be flood cooled. As a minimum, the coolant system shall consist of a reservoir, pump, strainer, and all necessary piping. The system shall be equipped with a method to control the rate of coolant flow from the reservoir and to direct coolant flow to the grinding area. A means shall be

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provided for flushing the catch pans, troughs, and other areas where swarf may accumulate. When specified (see 7.2(z)), an optional filtration system shall be provided.

3.2.15 Hydraulic system. If provided, the hydraulic system shall conform to the requirements of National Fluid Power Association (NFPA) T2.24.1. The hydraulic system shall be complete and include all pumps, valves, piping, cylinders, and controls. Means shall be provided to prevent damage to the components from overpressure. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.2.16 Electrical system. Unless otherwise specified (see 7.2(aa)), the electrical system shall conform to the National Fire Protection Association (ANSI/NFPA) 79. The machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(bb)). Unless otherwise specified (see 7.2(cc)), the control system power shall be provided by a separate 120-volt, single-phase, 60-hertz circuit from the machine main power supply.

3.2.17 Electric motors. The grinding machines drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1. Unless otherwise specified (see 7.2(dd)), the motors shall have open dripproof enclosures.

3.2.18 Optional accessories. Optional accessories shall be provided as specified (see 7.2(ee)).

3.3 Performance. Unless otherwise specified (see 7.2(ff)), the grinding machine shall meet the alignment requirements shown in table III and the performance requirements shown in table IV when tested in accordance with sections 5.3 through 5.3.5.

TABLE III. Alignment requirements.

Parameter	Tolerance
Worktable flatness: deviation from mean	±0.0001 inch
Worktable travel: rise or fall, over full travel	±0.0001 inch
Worktable surface: parallel with Z-axis, over full travel	±0.0001 inch
Wheel spindle, axial slip and radial runout: up to 12.1 horsepower over 12.1 horsepower	0.0001 inch TIR[1] 0.0003 inch TIR
Squareness of axes: X- and Z-axes shall be square to each other, and the Y-axis shall be square to the plane described by X and Z	0.001 inch/foot TIR
Machineway pairs: surfaces flat and straight, edges parallel	±0.0001 inch/foot

[1] TIR is total indicator readout.

TABLE IV. Performance requirements.

Parameter	Tolerance
Workpiece: surfaces flat	±0.0001 inch/foot
Workpiece: surfaces parallel	±0.0001 inch/foot
Workpiece: finished surface, Rt[1]	8 microinches
Workpiece: finished dimensions	±0.0001 inch

[1] Rt is roughness total as defined in American Society of Mechanical Engineers (ASME) B46.1.

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3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(gg)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.9. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(hh)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(ii)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(jj)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
h. National stock number.
i. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.2(kk)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the

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emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The grinding machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(ll)), the grinding machine shall be tested in accordance with sections 5.3.1 through 5.3.5.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(mm)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(nn)). Dimensional measurements shall be validated and compensated for in accordance with environmental test conditions and methods specified in American Society of Mechanical Engineers (ASME) B89.6.2.

5.3.2 Operational test. The grinding machine shall be operated in accordance with the manufacturer's standard operating procedures. The grinding machine shall be operated at no load for not less than 1 hour. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Alignment tests. After the operational test, the alignment shall be verified before and after the performance tests and shall meet the requirements shown in table III. Changes in alignment shall be cause for rejection.

5.3.4 Performance tests. The performance tests shall be conducted in accordance with the manufacturers standard operating procedures. The grinding machine shall have been operated for at least two hours and reached a stable operating temperature as a requisite for the tests.

5.3.4.1 Block grinding test. A minimum of five test blocks shall be ground top and bottom to produce flat and parallel surfaces. One block shall be located at each corner and one block shall be located in the center of the worktable. On worktables up to 36 inches long and/or 18 inches wide, the blocks shall be at least 1 inch thick and shall present a surface area of at least 4 square inches on each face to be ground. On worktables greater than 36 inches long and 18 inches wide, the block surface area to be ground shall be equal to 9 square inches. Additional blocks shall be added to the worktable as required to maintain the distance between blocks at 2 feet or less.

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5.3.4.2 Surface finish grinding test. The test blocks prepared by the block grinding test shall be finish ground on both top and bottom surfaces. The grinding wheel may be changed prior to but not during the finish grind. The total roughness (Rt) shall meet the requirements shown in table IV.

5.3.5 Optional tests. Optional tests shall be conducted as specified (see 7.2(oo)).

5.4 Acceptance. Unless otherwise specified (see 7.2(pp)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment and performance tests shall be compared with the requirements shown in tables III and IV, respectively. Failure of the machine to meet these requirements shall be cause for rejection.

6. PACKAGING. Preservation, packing, and marking shall be as specified (see 7.2(qq)).

7. NOTES

7.1 Sources of referenced documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

American National Standards Institute (ANSI)

ANSI B5.1	T-Slots, Their Bolts, Nuts, and Tongues
ANSI B7.1	Safety Requirements for the Use, Care and Protection of Abrasive Wheels
ANSI B11.9	Safety Requirements for the Construction, Care, and Use of Grinding Machines
ANSI B74.13	Markings for Identifying Grinding Wheels and Other Bonded Abrasives

Applications for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.
American Society of Mechanical Engineers (ASME)

ASME B46.1	Classification and Designation of Surface Qualities
ASME B89.6.2	Temperature and Humidity Environment for Dimensional Measurement

Applications for copies should be sent to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

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Electronic Industries Association (EIA)

ANSI/EIA 267-C	Axis and Motion Nomenclature for Numerically Controlled Machine

Applications for copies should be sent to Electronic Industries Association, 2500 Wilson Boulevard, Arlington, VA 22201-3834.

National Electrical Manufacturers Association (NEMA)

NEMA MG-1	Motors and Generators

Applications for copies should be sent to the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

National Fire Protection Association (NFPA)

ANSI/NFPA 79	Electrical Standards for Industrial Machinery

Applications for copies should be sent to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.

National Fluid Power Association (NFPA)

ANSI/NFPA/JIC T2.24.1	Hydraulic Fluid Power - Systems Standard for Stationary Industrial Machinery

Applications for copies should be sent to the National Fluid Power Association, 3333 N. Mayfair Road, Milwaukee, WI 53222-3219.

Occupational Safety and Health Administration (OSHA)

OSHA 29 PART 1910	Occupational Safety and Health Standards for General Industry (29 CFR PART 1910)

Applications for copies should be sent to the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

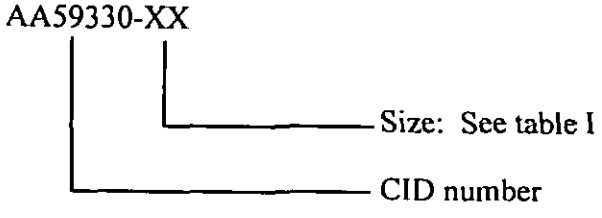
7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Size grinding machine required (see 2).
c. Machineways, if different (see 3.2.2).
d. Table travel, if different (see 3.2.3).
e. Mounting means on worktable for fixture/chuck, if different (see 3.2.3).
f. Number and size of T-slots, as specified (see 3.2.3).
g. Magnetic chuck, as specified (see 3.2.4).
h. Magnetic chuck surface area, if different (see 3.2.4).
i. Spindle drive, horsepower, and speed, if different (see 3.2.6).

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j. Spindle for premounted wheels, as specified (see 3.2.6).
k. Spindle height, if different (see 3.2.6).
l. Grinding wheel(s), if different (see 3.2.7).
m. Grinding wheel dresser, if different (see 3.2.8).
n. Grinding wheel forming device, if required (see 3.2.9).
o. Graduations, if different (see 3.2.10).
p. Digital readout, if required (see 3.2.10.1).
q. Digital readout requirements, if different (see 3.2.10.1).
r. Control panel, if different (see 3.2.11.1).
s. Handwheel feed controls, if different (see 3.2.11.2).
t. Power feed controls, if different (see 3.2.11.3).
u. Automatic down feed cycle control, if required (see 3.2.11.4).
v. Dust extract system, if different (see 3.2.12).
w. Central extractor interface, as specified (see 3.2.12).
x. Automatic lubrication, if required (see 3.2.13).
y. Coolant system, if different (see 3.2.14).
z. Filtration system, if required (3.2.14).
aa. Electrical system industrial standard, if different (see 3.2.16).
bb. Electrical system voltage, as specified (see 3.2.16).
cc. Control system voltage, if different (see 3.2.16).
dd. Motor enclosure, if different (see 3.2.17).
ee. Optional accessories, as specified (see 3.2.18).
ff. Performance, if different (see 3.3).
gg. Machine dimensions and weight, as specified (see 3.4).
hh. System of units required (see 3.6).
ii. Dual system of units, if required (see 3.6).
jj. Nameplate, if different (see 3.7).
kk. Lubrication plate or chart, if required (see 3.8).
ll. Testing, if different (see 5.3).
mm. Test responsibility, if different (see 5.3.1).
nn. Certification of calibration, as specified (see 5.3.1).
oo. Optional tests, as specified (see 5.3.5).
pp. Acceptance, if different (see 5.4).
qq. Packaging requirements, as specified (see 6).

7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



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MILITARY INTERESTS:

Custodians
Air Force - 99
Army - AL
Navy - SH

Reviewers
Air Force - 84
Army - AR, SM
Navy - MC

PREPARING ACTIVITY:
DLA - GS

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